CJ Nour

Electrical/Software Engineer
Residing in Etobicoke, Ontario

Portfolio: cjnour.com
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SKILLS

Electrical: EPLAN, PCB Design, Soldering, Servo Motor Control, PLC Programming
Software: Java, React.js, Python, C++, Git, Linux
CAD: NX, Autodesk Inventor, EPLAN, 3D Printing & Prototyping
Analysis: Excel, MATLAB/Simulink, Root Cause Analysis, Hardware-in-the-Loop Testing

PROFESSIONAL EXPERIENCE

Husky Technologies | Electrical Engineer — Jun. 2025 – Present
EPLAN, Siemens NX, UL 508A, Root Cause Analysis — *Bolton, Ontario*

- Leading the electrical design of prototypes for a new machine generation. Creating schematics in EPLAN, CAD in Siemens NX, sizing cable assemblies, and validating grounding to UL 508A. Diagnosed and resolved a 9 year-old service issue related to Ethernet cable routing, cutting critical communications faults by 100%.
- Resolved 30+ non-conformances on a new machine family (UL & EN builds; CE and non-CE cabinets). Corrected panel schematic errors critical to safety devices, fixed assembly drawing errors. Achieved full release compliance by eliminating 100% of the machine's electrical cabinets release blockers.
- Authoured and published three enterprise-level electrical standards to drive proactive refresh of data and reduce engineering errors. Presented to engineering leadership to secure adoption across disciplines to actively sustain a multi-million dollar per year business.

Tesla | Controls Engineer Intern — Jun. 2025 – Present
EPLAN, Siemens NX, UL 508A, Root Cause Analysis — *Markham, Ontario*

- Spearheaded the development of an RFID-based monitoring system that aims to reduce tooling maintenance issues by over 90%. Integrated PLC logic and developed a custom TwinCAT HMI dashboard to track service history and collect tooling lifespan data on multiple Gigafactory assembly lines.
- Contributed to commissioning 3 automation machines by leveraging Beckhoff TwinCAT3 Structured Text to program critical components, including VFDs, sensors, and robotic and pneumatic devices. Debugged field I/O networks using multiple industrial communication protocols (TCP/IP, EtherCAT, Profinet).
- Led multiple R&D initiatives to solve critical engineering challenges to enhance Beckhoff PLC and FANUC robot performance by experimenting with computer vision systems, time-of-flight lasers, profilometers. Authoured test result reports and applied technical specifications to optimize device integration.

Material Planner Intern — Sep. 2023 – Dec. 2023

- Developed and sustained 3 Excel VBA tools to optimize inventory and logistics, eliminating duplicate shipments by 100%. Automated the parts reordering process, automated gap analyses for project needs, and streamlined shipping with auto-generated forms and emails - enhancing database operational efficiency.
- Leveraged these tools to identify inefficiencies in the existing third-party inventory system and built a business case to influence executives to create an internal software team to develop an improved, proprietary system.
- Facilitated international shipments, expedited missing parts, and streamlined kitting to reduce technician downtime by 20%, ensuring efficient workflows across Gigafactories and optimizing daily resource allocation.

Moneris | Software Engineer Intern — Jun. 2022 – Apr. 2023
Change me — *Toronto, Ontario*

- Developed a tool boosting productivity for 30+ employees by enabling credit card transaction analytics. Built with JavaScript and React, the tool allows users to customize dashboards and manage widgets seamlessly.
- Improved code efficiency by 50% by implementing Redux and a backend system hosted on Parse, optimizing personal and team tools by implementing reducers, utilizing dispatch, and querying user database properties.
- Collaborated in agile feedback sessions with the Scrum Master, QA team, Architecture team, and other stakeholders to identify and resolve platform issues related to our proprietary payment platform testing tool.

Projects

Autonomous Defense Drone System

- Developing an autonomous defense drone system to secure airspace by detecting, tracking, and intercepting unauthorized drones. The system integrates **multi-camera detection**, robotic arms designed in **CAD**, and an autonomous drone to provide a solution to drone-related threats.
- Utilizing **RF Tx/Rx** modules to enable communication between the camera hub system and the drones. Implementing **YOLOv8**, an **AI camera**, and **control algorithms** in **Python** to command the motors and autonomously track, pursue, and intercept unauthorized drones.

Autonomous RC Car

- Implemented self-driving algorithms coded in **C++ and Python** on an RC car to navigate through obstacles via **LiDAR** and **ultrasonic sensor** data processed through a **NVIDIA Jetson Nano** running on **Linux Ubuntu**.
- Coordinated **agile** team meetings as the Scrum Master to monitor progress, assign tasks, address challenges, and troubleshoot issues during development and testing phases.

Automated Vanity System

- Designed and fabricated a custom automated vanity system from a recycled oil barrel. Cut and welded a panel from the barrel to a recycled chair and added wheels for smooth accessibility.
- Engineered a motorized mirror and a **Bluetooth** sound system by wiring a recycled car window regulator motor to a momentary rocker switch and a power supply with an **AC/DC rectifier**. Researched **torque-speed characteristics** and motor specifications to ensure the motor could support the load with the supplied power.

Education

McMaster University | B.Eng., Electrical Engineering — Graduated Apr. 2025